RCA Interview Practice — Solutions

Model answers, interviewer reveals and common mistakes for all 24 scenarios

24 root-cause-analysis scenarios · Easy / Medium / Hard

12 November 2026

Independent practice material. All scenarios are hypothetical; company names only identify the type of product. Not affiliated with or endorsed by any company named.

How to use the solutions

If you are the practice interviewer, read the **Interviewer reveals** before the session and share each fact only when the candidate asks a question that would uncover it. If you are self-studying, attempt the question first, then compare your path with the model path below.

Each model answer shows one strong path to the root cause. Other orders of investigation are fine if they are structured and prioritised. All facts are invented for practice.

Q1 · Cart additions down 15% in three days

E-commerce marketplace (Flipkart-style) · Metric: Add-to-cart events · Medium

Prompt

Cart additions on a large e-commerce marketplace have fallen about 15% over the last three days. Diagnose the problem.

1. Clarifying questions to ask

- What exactly counts as a cart addition: a tap on Add to cart, or the item successfully appearing in the cart?
- Is the 15% across web, iOS and Android, or concentrated on one surface?
- Sudden step change or gradual decline? Compared with what baseline (same days last week)?
- Has the data pipeline or event definition changed?

2. Hypothesis buckets

- Data issue: event renamed, SDK change, delayed pipeline.
- Internal change: new app release, A/B test, pricing or catalogue change, outage.
- External: competitor sale, holiday, payment-network issue, news.
- Segment-specific: one platform, version, region, category or user type.

Interviewer reveals (share only when asked)

- The drop is almost entirely on the mobile app.
- Analytics logging has been verified as healthy.
- No change by city, age or gender.
- A new app version shipped four days ago that reworked the cart and checkout flow.
- Android only. Android cart additions on the new version are close to zero; iOS is normal.
- A few new bug reports mention being unable to check out.

3. Model investigation path

1. Confirm data quality — logging verified, so the drop is real.
2. Segment by surface: concentrated on mobile app.
3. Rule out demographics and location: drop is uniform, which points away from market causes.
4. Check recent releases: a cart/checkout rework shipped four days ago.
5. Split by OS and app version: Android new version shows near-zero additions; iOS normal.
6. Walk the funnel on the affected version: product page views normal, Add-to-cart taps normal, cart-add confirmations missing.

Root cause

A bug in the new Android release breaks the step between tapping Add to cart and the item being saved to the cart (for example a failing API call or a crashed handler). The 15% overall drop is the Android share of traffic hitting zero conversions on that step.

FIX

- Immediately: roll back or kill-switch the new cart flow on Android; push a hotfix; consider forcing an update.
- Communicate with support so agents can explain and help affected users.
- Estimate lost orders and revenue for leadership.

MEASURE RECOVERY

- Primary: add-to-cart success rate by OS and app version returns to baseline.
- Secondary: checkout starts and orders on Android.
- Guardrail: crash rate, app-store rating, support tickets.

PREVENT RECURRENCE

- Staged rollout (1% → 10% → 50% → 100%) with automatic alerts on funnel-step conversion by version.
- End-to-end tests for the add-to-cart flow on real Android devices before release.

4. Common mistakes

- Spending ten minutes on competitors before checking internal releases.
- Not asking about platform and app version early.
- Proposing a redesign instead of finding the broken step.

60-second answer

Confirmed the data was real, segmented by surface and found the drop on mobile only, ruled out demographic and market causes because the drop was uniform, then tied it to a release four days ago. Splitting by OS showed Android's new version had near-zero cart additions while taps were normal, so the break is between the tap and the cart save. Roll back, hotfix, and add staged rollouts with per-version funnel alerts.

Q2 · Food-delivery orders down 10% in one city

Food delivery (Swiggy/Zomato-style) · Metric: Completed orders · Medium

Prompt

Completed orders in Bengaluru fell 10% this week versus last week. Other cities are flat. What happened?

1. Clarifying questions to ask

- Completed orders or placed orders? Week over week, same days?
- All of Bengaluru or certain zones? All hours or peak hours?
- Any change to fees, incentives, restaurants or the app this week?

2. Hypothesis buckets

- Demand: weather, local holiday, competitor promotion.
- Supply: restaurants offline, rider shortage.
- Product: payment or address bug for that region.
- Policy: fee or incentive change.

Interviewer reveals (share only when asked)

- App sessions in Bengaluru are flat.
- Menu views and carts are flat, but checkout-to-completion fell.
- Order cancellations by the platform rose sharply in the evening.
- Rider log-ins in Bengaluru fell 20% after a rider-incentive change on Monday.

3. Model investigation path

1. Sessions flat → demand is steady.
2. Funnel stable until after order placement → issue is fulfilment, not intent.
3. Platform cancellations up at dinner peak → no rider assigned in time.
4. Rider supply down 20% after Monday's incentive change.

Root cause

A change to rider incentives in Bengaluru reduced rider log-ins at peak, so orders could not be assigned and were auto-cancelled.

FIX

- Restore or adjust peak incentives; temporarily raise delivery radius or batching.
- Show honest ETAs and pause ordering in zones with no riders rather than cancelling after payment.

MEASURE RECOVERY

- Rider log-ins at peak, assignment time, platform-cancellation rate, completed orders.
- Guardrails: rider earnings per hour, cost per order.

PREVENT RECURRENCE

- Test incentive changes in a few zones first; alert on assignment time and cancellations.

4. Common mistakes

- Assuming a demand problem without checking supply.
- Ignoring the two-sided nature of the marketplace.

60-second answer

Demand was steady and the funnel was fine until after orders were placed. Platform cancellations jumped at dinner time because riders weren't available: rider log-ins dropped 20% after an incentive change. Restore peak incentives, stop taking orders we can't fulfil, and pilot future incentive changes by zone.

Q3 · Ride cancellations up 25%

Ride-hailing (Uber/Ola-style) · Metric: Cancellation rate · Medium

Prompt

Cancellation rate across the platform rose from 12% to 15% over two weeks. Find the cause.

1. Clarifying questions to ask

- Rate defined as cancelled ÷ requests? Rider- or driver-initiated?
- All cities or some? All trip types?
- Any pricing, matching or app changes in the period?

2. Hypothesis buckets

- Supply-demand imbalance (events, rain).
- ETA inaccuracy causing rider cancellations.
- Driver economics: short trips, long pickups, unattractive destinations.
- App change exposing new information at a new moment.

Interviewer reveals (share only when asked)

- Driver-initiated cancellations explain most of the rise.
- They cluster after the driver sees the drop-off location.
- A product change two weeks ago added a 'trip details' screen that shows drivers the destination right after they accept.
- Long airport-bound trips from city edges see the highest cancellations.

3. Model investigation path

1. Split by canceller → drivers drive the increase.
2. Split by timing → after acceptance, when trip details appear.
3. Correlate with release → new trip-details screen shows destination after acceptance.
4. Segment by trip → long pickups and out-of-zone drop-offs most affected.

Root cause

A new screen shows drivers the destination after they accept; drivers now cancel trips they find unattractive, which they previously completed.

FIX

- Short term: revert the timing or add a small fee for post-accept cancellations.
- Longer term: fix economics of unattractive trips (return-trip bonus, long-pickup pay) rather than hiding information.

MEASURE RECOVERY

- Driver post-accept cancellation rate, rider wait time, completed trips per hour.
- Guardrails: driver earnings, rider complaints, fairness across areas.

PREVENT RECURRENCE

- Model behavioural impact of exposing new information; launch with holdouts.

4. Common mistakes

- Treating cancellation as a single metric instead of splitting by who and when.

60-second answer

The rise was driver cancellations right after accepting, when a new screen revealed the destination. Drivers dropped trips they found unattractive. Revert the timing now, then fix the underlying trip economics and measure post-accept cancellations, wait times and driver earnings.

Q4 · Payment success rate fell from 95% to 91%

Payments gateway (Razorpay/Stripe-style) · Metric: Payment success rate · Hard

Prompt

Payment success rate for merchants dropped from 95% to 91% yesterday afternoon. Diagnose.

1. Clarifying questions to ask

- Success = successful ÷ attempted? Excluding user-abandoned?
- All merchants or some? All methods?
- Sudden or gradual? Still ongoing?

2. Hypothesis buckets

- Our deploy or config change.
- Partner outage (bank, card network, UPI switch).
- Fraud rule change blocking legitimate payments.
- Merchant integration issue.

Interviewer reveals (share only when asked)

- Cards and wallets are normal; UPI success fell from 96% to 84%.
- Failures are concentrated on two large issuer banks.
- Error codes are mostly timeouts from the bank side.
- No deploy on our side; one bank posted a maintenance notice late.

3. Model investigation path

1. Split by method → UPI only.
2. Split by issuer → two banks.
3. Error codes → bank-side timeouts.
4. Check deploys and partner status → bank maintenance window.

Root cause

Unannounced maintenance at two large issuer banks caused UPI timeouts; our system kept routing those transactions without fallback.

FIX	MEASURE RECOVERY	PREVENT RECURRENCE

- Route around unhealthy banks where possible; suggest alternate methods to customers; clear status-page communication to merchants.
- Ensure idempotent retries so customers aren't double charged.

- Success rate by method and issuer, timeout rate, duplicate-charge reports.

- Real-time issuer health monitoring with automatic routing and alerts; partner escalation contacts.

4. Common mistakes

- Assuming our code broke before checking partners.
- Ignoring duplicate charges during recovery.

> **60-second answer**
>
> Breaking the rate down showed only UPI failed, for two banks, with bank-side timeouts and no deploy from us. A bank maintenance window was the cause. Route around unhealthy banks, guide customers to other methods, keep retries idempotent, and build issuer-health monitoring with automatic routing.

Q5 · Credit-card application completions down 20%

Card issuer (American Express-style) · Metric: Applications submitted ÷ started · Medium

Prompt

Digital credit-card application completions dropped 20% this month. Find out why.

1. Clarifying questions to ask

- Is it completions in count or completion rate?
- Which channels, devices and steps?
- Any form, policy or vendor changes?

2. Hypothesis buckets

- Form or KYC vendor change.
- Mix shift from new acquisition channel.
- Eligibility or credit policy change.
- Technical bug on a device.

Interviewer reveals (share only when asked)

- Starts are up 30% thanks to a new paid-social campaign.
- Completion rate is down for new paid-social traffic only; organic users are normal.
- Paid-social applicants abandon at the income and document step.

3. Model investigation path

1. Split rate vs volume → rate down, volume up.
2. Split by channel → only new paid-social traffic converts poorly.
3. Funnel by step → those users drop at income/documents, suggesting lower eligibility or intent.

Root cause

Mix shift: a new paid-social campaign brought many lower-intent or ineligible applicants who abandon at the income and document step. The product itself is not broken.

FIX

- Retarget the campaign; add a pre-eligibility check before

MEASURE RECOVERY

- Completion and approval rate by channel; cost per

PREVENT RECURRENCE

- Report funnel metrics segmented by acquisition

the long form.
- Judge channels on approved and activated accounts, not starts.

approved account.

channel by default.

4. Common mistakes

- Redesigning the form when the traffic changed, not the product.

> **60-second answer**
>
> Starts rose 30% from a new paid-social campaign, but only those users completed less, dropping at income and documents. It's a mix shift, not a product bug. Retarget the campaign, add a pre-eligibility step, and measure channels by approved, activated accounts.

Q6 · Daily active users down 8% on a social app

Social media app · Metric: DAU · Medium

Prompt

DAU fell 8% over the past week. Leadership wants to know why.

1. Clarifying questions to ask

- How is 'active' defined? Same definition as before?
- Which platforms, regions, cohorts?

2. Hypothesis buckets

- Seasonality or holiday.
- Notification delivery issue.
- Feed quality or ranking change.
- Competitor launch.

Interviewer reveals (share only when asked)

- New user sign-ups are flat; returning user DAU is down.
- Push notification open rate fell sharply.
- Android 14 users received far fewer notifications after an OS permission change was handled incorrectly in our latest release.

3. Model investigation path

1. Decompose DAU → returning users down.
2. Main entry point for returning users is notifications → open rate dropped.
3. Split by OS version → Android 14 deliveries down after release.

Root cause

The latest release mishandled the Android notification-permission prompt, so many Android 14 users stopped receiving notifications and opened the app less.

FIX

- Hotfix the permission flow; re-request permission in

MEASURE RECOVERY

- Notification delivery and open rate by OS version;

PREVENT RECURRENCE

- Monitor notification delivery by OS version as a release

context with a clear reason.

returning DAU.
- Guardrails: opt-out rate, uninstalls.

health metric.

4. Common mistakes

- Blaming content or competitors without decomposing DAU.

60-second answer

Returning users fell, not new ones. Their main trigger, notifications, collapsed on Android 14 after our release mishandled the permission prompt. Hotfix and re-ask permission in context, and add notification delivery by OS version to release monitoring.

Q7 · Streaming watch time down 12%

Video streaming (Netflix-style) · Metric: Hours watched per member · Medium

Prompt

Average weekly watch time per member fell 12% month over month. Diagnose.

1. Clarifying questions to ask

- Per member or total? Same comparison period?
- Any playback, quality or app changes?

2. Hypothesis buckets

- Content seasonality.
- Playback quality issues causing early exits.
- Recommendation change.
- Competitor content.

Interviewer reveals (share only when asked)

- Sessions per member are steady; average session length fell.
- Last month had the finale of a very popular series; this month has no big release.
- Drop is across all regions and devices.

3. Model investigation path

1. Uniform across segments → not a device or region bug.
2. Shorter sessions, not fewer → users still come back.
3. Compare with content calendar → last month had a major finale.

Root cause

Natural seasonality: last month's spike from a major series finale inflated the baseline; this month returns to normal levels.

FIX

- No product fix needed; compare against a seasonally

MEASURE RECOVERY

- Watch time vs seasonal baseline; churn among finale watchers.

PREVENT RECURRENCE

- Annotate dashboards with content releases.

adjusted baseline or same period last year.
- Consider surfacing back-catalogue recommendations after big finales.

4. Common mistakes

- Inventing a bug when the baseline was unusually high.

60-second answer

The drop is uniform and session counts are steady, only length fell. Last month had a huge finale that inflated the baseline. This is seasonality; compare against a seasonal baseline and use the moment to promote back-catalogue titles.

Q8 · App store rating dropped from 4.5 to 3.9

Fintech app · Metric: Average store rating · Easy

Prompt

The app's rating dropped from 4.5 to 3.9 in two weeks. Why?

1. Clarifying questions to ask

- Which stores? Which versions? How many new reviews?
- Did we change in-app rating prompts?

2. Hypothesis buckets

- Bug in a release.
- Fee or policy change.
- Change in who is prompted to rate.

Interviewer reveals (share only when asked)

- Most new 1-star reviews mention being logged out repeatedly.
- They started after version 6.2, which changed session handling.
- Crash rate is normal.

3. Model investigation path

1. Tag reviews → logouts dominate.
2. Correlate with version → started at 6.2.
3. Release notes → session-timeout change.

Root cause

Version 6.2 shortened session timeouts, forcing frequent re-logins and frustrating users.

FIX

- Restore a reasonable timeout with biometric re-authentication; respond to reviewers once fixed.

MEASURE RECOVERY

- Login events per user, rating trend, review themes.

PREVENT RECURRENCE

- Track re-authentication frequency as a release metric; involve security and UX together.

4. Common mistakes

- Responding to reviews without diagnosing the cause.

> **60-second answer**
>
> Tagging the reviews showed repeated logouts starting with version 6.2, which shortened session timeouts. Restore a sensible timeout with biometric re-login, reply to reviewers, and track re-login frequency per release.

Q9 · Search-to-purchase conversion down 9%

E-commerce search · Metric: Purchases from search ÷ searches · Hard

Prompt

Conversion from search to purchase fell 9% over a week. Diagnose.

1. Clarifying questions to ask

- Conversion per search or per searcher? Same window?
- Which languages, categories, platforms?

2. Hypothesis buckets

- Ranking or index change.
- Catalogue gaps (out of stock).
- Price changes.
- Language or spelling handling.

Interviewer reveals (share only when asked)

- Search volume is flat. Click-through on results fell; zero-result searches doubled.
- Zero-result queries are mostly in Hindi and Hinglish.
- A new tokenizer was deployed to the search service last Tuesday.

3. Model investigation path

1. Search volume flat, CTR down → results got worse.
2. Zero-result rate doubled → index or query processing.
3. Segment by language → Hindi/Hinglish.
4. Correlate with deploy → new tokenizer.

Root cause

A new tokenizer broke handling of Hindi and transliterated queries, returning zero or poor results.

FIX

- Roll back the tokenizer for those languages; add

MEASURE RECOVERY

- Zero-result rate, CTR and conversion by language.

PREVENT RECURRENCE

- Offline search evaluation set covering languages and

multilingual test queries.

spelling variants before every ranking deploy.

4. Common mistakes

- Looking only at overall conversion, not intermediate funnel metrics like CTR and zero-results.

60-second answer

Searches were flat but click-through fell and zero-result searches doubled, mostly for Hindi and Hinglish queries, right after a tokenizer deploy. Roll it back for those languages and require a multilingual evaluation set before search changes.

Q10 · Revenue down while DAU is up

Freemium productivity app · Metric: Monthly revenue · Hard

Prompt

DAU grew 10% this month but subscription revenue fell 5%. Explain.

1. Clarifying questions to ask

- Revenue recognised or billed? Gross or net of refunds?
- Which plans and regions?

2. Hypothesis buckets

- Mix shift to free users.
- Payment failures (involuntary churn).
- Price or discount changes.
- Refunds.

Interviewer reveals (share only when asked)

- New users are mostly from a student promotion and are on the free tier.
- Existing paid users: renewal failures rose because card payments failed after a billing provider change.

3. Model investigation path

1. Decompose → payer conversion down (mix shift), and revenue per payer down.
2. Renewal failure rate up → billing provider change broke card-on-file renewals.

Root cause

Two causes: a student promotion added many free users (diluting conversion, which is fine), and a billing-provider migration broke card renewals (involuntary churn, which is not).

FIX

- Fix renewal billing; run dunning (retries, card update emails) for failed renewals.

MEASURE RECOVERY

- Renewal success rate, involuntary churn, revenue per payer.

PREVENT RECURRENCE

- Migration checklists with shadow billing runs before cut-over.

- Report free and paid cohorts separately.

4. Common mistakes

- Stopping at the first explanation (mix shift) and missing the payment bug.

60-second answer

Revenue = users × payer rate × revenue per payer. DAU rose from free student sign-ups, which is expected dilution. But renewals were also failing after a billing-provider migration. Fix renewals and run dunning, and always look for a second cause.

Q11 · Average order value down 7%

Quick commerce (Blinkit-style) · Metric: AOV · Medium

Prompt

Average order value fell 7% this month. Investigate.

1. Clarifying questions to ask

- AOV including fees and discounts? Same definition?
- Which cities and categories?

2. Hypothesis buckets

- Threshold or fee change.
- Category mix shift.
- Discounting.
- Recommendation module broken.

Interviewer reveals (share only when asked)

- Price per item is flat; items per order fell.
- Free-delivery threshold was lowered from ₹199 to ₹99 two weeks ago.
- Order count rose 12%.

3. Model investigation path

1. Decompose AOV → items per order down.
2. Check pricing changes → lower free-delivery threshold.
3. Order count up → users split baskets into smaller, more frequent orders.

Root cause

Lowering the free-delivery threshold let users place smaller, more frequent orders. AOV fell but order count rose.

FIX

- Decide on contribution margin, not AOV: if delivery

MEASURE RECOVERY

- Contribution margin per order and per user, order frequency, delivery cost.

PREVENT RECURRENCE

- Evaluate fee changes on unit economics and total GMV per user, not one metric.

cost per order now exceeds gains, adjust the threshold.

4. Common mistakes

- Calling lower AOV a problem without checking total revenue and margin.

60-second answer

AOV dropped because items per order fell after the free-delivery threshold was lowered. Orders went up 12%. Whether this is bad depends on contribution margin per user, so judge the change on unit economics, not AOV alone.

Q12 · Sign-ups down 30% overnight

Consumer app · Metric: New sign-ups · Easy

Prompt

New sign-ups fell 30% overnight. What do you check?

1. Clarifying questions to ask

- Sign-ups started or completed? All platforms?

2. Hypothesis buckets

- Tracking break.
- Sign-up flow bug.
- OTP/SMS delivery failure.
- Acquisition campaign paused.

Interviewer reveals (share only when asked)

- Installs are normal.
- OTP delivery success fell from 98% to 70%.
- The SMS provider had an incident in one region.

3. Model investigation path

1. Installs normal → acquisition is fine.
2. Funnel → users stuck at OTP verification.
3. OTP delivery down → SMS provider incident.

Root cause

An SMS provider incident stopped OTP delivery in one region, blocking sign-up completion.

FIX

- Fail over to a secondary SMS provider; offer voice OTP or email fallback.

MEASURE RECOVERY

- OTP delivery rate by provider and region; sign-up completion.

PREVENT RECURRENCE

- Multi-provider OTP with automatic failover and alerts.

4. Common mistakes

- Pausing marketing spend before checking the funnel.

60-second answer

Installs were normal but users got stuck at OTP verification because an SMS provider failed in one region. Fail over to a backup provider, add voice or email fallback, and alert on OTP delivery by provider.

Q13 · Support tickets doubled after a release

SaaS project-management tool · Metric: Support tickets per 1,000 users · Easy

Prompt

Support tickets doubled in the week after a release. Diagnose.

1. Clarifying questions to ask

- Which ticket categories? All customer segments?

2. Hypothesis buckets

- Bug.
- Discoverability problem from UI change.
- Pricing change.

Interviewer reveals (share only when asked)

- Most tickets ask where the export button went.
- The redesign moved export into a new 'More' menu.

3. Model investigation path

1. Tag tickets → 'can't find export' dominates.
2. Release notes → export moved into overflow menu.

Root cause

A redesign hid a frequently used action, creating a discoverability problem rather than a bug.

FIX

- Restore export as a visible action or add an in-app pointer; update help articles.

MEASURE RECOVERY

- Ticket volume by theme, export usage.

PREVENT RECURRENCE

- Usability-test redesigns on top tasks; check feature usage before moving controls.

4. Common mistakes

- Treating tickets as noise because nothing is 'broken'.

60-second answer

Tickets doubled mainly because users couldn't find export after the redesign moved it into a menu. Make export visible again, add a pointer and help update, and usability-test top tasks before moving controls.

Q14 · Checkout conversion spiked 20%

E-commerce · Metric: Checkout conversion · Medium

Prompt

Checkout conversion jumped 20% overnight. Celebrate or investigate?

1. Clarifying questions to ask

- Is the numerator or denominator driving it?

2. Hypothesis buckets

- Real improvement (promo, UX).
- Denominator dropped (tracking or bot filter).
- Mix shift.

Interviewer reveals (share only when asked)

- Orders didn't change; checkout starts fell.
- A tracking change stopped firing checkout_start for guest users.

3. Model investigation path

1. Numerator (orders) flat → no real gain.
2. Denominator down → checkout_start events missing.
3. Tracking release → guest event broken.

Root cause

A tracking bug stopped logging checkout starts for guest users, shrinking the denominator and inflating conversion.

FIX

- Fix the event; annotate the dashboard; restate affected numbers.

MEASURE RECOVERY

- Reconcile events with backend order and session data.

PREVENT RECURRENCE

- Automated checks comparing event counts with backend counts.

4. Common mistakes

- Celebrating before checking numerator and denominator separately.

> **60-second answer**
>
> Orders were flat; checkout starts dropped because guest checkout_start events stopped firing. The 'improvement' was a tracking bug. Fix the event, restate numbers, and reconcile analytics with backend counts automatically.

Q15 · Delivery times up 6 minutes

Food delivery · Metric: Average delivery time · Medium

Prompt

Average delivery time increased by six minutes this week. Why?

1. Clarifying questions to ask

- Mean or median? All cities?

2. Hypothesis buckets

- Traffic or weather.
- Rider shortage.
- Restaurant prep delays.
- Dispatch algorithm change.

Interviewer reveals (share only when asked)

- Rider travel time is unchanged.
- Time waiting at restaurants rose.
- A new 'restaurant ready' estimate went live, assigning riders earlier.

3. Model investigation path

1. Decompose: prep + wait + travel.
2. Wait at restaurant up; travel flat.
3. New dispatch model assigns riders too early, so they queue at restaurants and lose other trips.

Root cause

A dispatch change assigned riders too early, so they waited at restaurants instead of delivering other orders, increasing overall delivery times.

FIX

- Roll back or recalibrate the ready-time model with live kitchen load.

MEASURE RECOVERY

- Rider wait time at restaurant, delivery time p50/p90, orders per rider-hour.

PREVENT RECURRENCE

- Launch dispatch changes as switchback experiments with

time and efficiency guardrails.

4. Common mistakes

- Blaming traffic without decomposing the delivery stages.

60-second answer

Travel was unchanged but riders waited longer at restaurants after a new readiness model assigned them too early. Recalibrate with live kitchen load and test dispatch changes with switchback experiments.

Q16 · Email open rate halved

B2C newsletter or lifecycle email · Metric: Open rate · Easy

Prompt

Open rate on lifecycle emails fell from 40% to 20%. Diagnose.

1. Clarifying questions to ask

- Open rate measured how? All email clients?

2. Hypothesis buckets

- Spam-folder deliverability problem.
- Subject line change.
- Measurement change (privacy features).

Interviewer reveals (share only when asked)

- Click-through rate per delivered email is unchanged.
- Apple Mail users show the drop; a privacy change altered open tracking.

3. Model investigation path

1. Clicks steady → engagement probably unchanged.
2. Split by client → Apple Mail only.
3. Measurement changed due to privacy features.

Root cause

A privacy feature change in one email client altered how opens are counted; real engagement did not fall.

FIX

- Move success metrics to clicks and conversions; note the change on dashboards.

MEASURE RECOVERY

- Click-through, conversions per send, unsubscribe rate.

PREVENT RECURRENCE

- Prefer robust downstream metrics over opens.

4. Common mistakes

- Rewriting emails because of a measurement artefact.

> **60-second answer**
>
> Clicks were steady and only one email client showed the drop, after a privacy change affected open tracking. Engagement didn't fall; switch to clicks and conversions as the primary metrics.

Q17 · Seller listings down 18%

Marketplace (seller side) · Metric: New listings per day · Medium

Prompt

New product listings by sellers dropped 18% in two weeks. Investigate.

1. Clarifying questions to ask

- Which categories and seller sizes?

2. Hypothesis buckets

- Seller churn.
- Listing tool bug.
- Policy or form change.
- Seasonality.

Interviewer reveals (share only when asked)

- Active sellers are steady; listings per seller fell.
- A new mandatory field (HSN tax code) was added to the listing form.
- Listing form abandonment doubled at that field.

3. Model investigation path

1. Sellers steady, listings per seller down.
2. Form funnel → abandonment at new mandatory field.

Root cause

A new mandatory tax field confused small sellers, who abandoned listings.

FIX

- Auto-suggest the code from category; allow saving drafts; add help text.

MEASURE RECOVERY

- Form completion rate, listings per seller, time to list.

PREVENT RECURRENCE

- Test form changes with small sellers; roll out gradually.

4. Common mistakes

- Assuming sellers are leaving the platform.

> **60-second answer**
>
> Sellers didn't leave; they abandoned listings at a new mandatory tax-code field. Auto-suggest the code, allow drafts, and test form changes with small sellers first.

Q18 · AI assistant escalations up 40%

AI customer-support assistant (fintech) · Metric: Escalation to human rate · Hard

Prompt

The share of AI support chats escalated to human agents rose from 25% to 35% this week. Diagnose.

1. Clarifying questions to ask

- Escalation by user request or by assistant? All intents?
- Any model, prompt or knowledge-base change?

2. Hypothesis buckets

- Model or prompt change.
- Retrieval returning stale or no documents.
- New question types (product launch).
- Bug in handoff logic.

Interviewer reveals (share only when asked)

- The increase is concentrated in fee-related questions.
- A new fee schedule was published but the knowledge base was not re-indexed.
- The assistant says 'I'm not sure' and escalates more on fee questions.

3. Model investigation path

1. Split by intent → fee questions.
2. Check retrieval → no current fee document found.
3. KB pipeline → new schedule not re-indexed.

Root cause

The fee schedule changed but the retrieval index was not refreshed, so the assistant couldn't find current fee information and correctly escalated.

FIX

- Re-index; add policy-owner workflow that triggers re-

MEASURE RECOVERY

- Escalation rate by intent, retrieval hit rate, answer

PREVENT RECURRENCE

- Freshness monitoring for key documents; run the

indexing on publication.

accuracy on eval set.

evaluation set after any knowledge change.

4. Common mistakes

- Blaming the model when retrieval data is stale.

60-second answer

Escalations rose for fee questions because a new fee schedule wasn't re-indexed, so retrieval found nothing current and the assistant handed off. Re-index, connect publishing to indexing, and monitor document freshness.

Q19 · Week-1 retention fell from 35% to 28%

Language-learning app · Metric: Week-1 retention · Medium

Prompt

Week-1 retention for new users dropped from 35% to 28% over the last three cohorts. Diagnose.

1. Clarifying questions to ask

- Retention defined as active in days 7–13? All acquisition channels?

2. Hypothesis buckets

- Acquisition mix shift.
- Onboarding change.
- Notification issue.
- Content difficulty.

Interviewer reveals (share only when asked)

- A new longer placement test was added to onboarding.
- Users who skip or fail the test retain much less.
- Lesson completion on day 1 fell.

3. Model investigation path

1. Channel mix stable.
2. Day-1 activation down.
3. New placement test → longer time to first lesson.

Root cause

A longer onboarding placement test delayed the first lesson and placed some users at too-hard levels, reducing activation and week-1 retention.

FIX

MEASURE RECOVERY

PREVENT RECURRENCE

- A/B test onboarding changes against activation metrics.

- Make the test optional or shorter; start with an easy first lesson.

- Time to first lesson, day-1 lesson completion, week-1 retention.

4. Common mistakes

- Looking at week-1 retention without examining day-1 activation.

60-second answer

Channel mix was stable but day-1 lesson completion fell after a longer placement test was added to onboarding. Shorten or make it optional, and A/B test onboarding changes on activation.

Q20 · Refund requests up 3x

E-commerce fashion · Metric: Refund requests · Medium

Prompt

Refund requests tripled this week for one category. Investigate.

1. Clarifying questions to ask

- Which category, sellers and regions?

2. Hypothesis buckets

- Quality issue with a seller.
- Wrong product images.
- Size chart change.
- Fraud.

Interviewer reveals (share only when asked)

- Return reason is mostly 'size doesn't fit'.
- A new size chart format was rolled out for that category.
- One major brand's chart used a different standard.

3. Model investigation path

1. Return reasons → sizing.
2. Category change → new size chart format.
3. Brand-level split → one brand's mapping wrong.

Root cause

A size-chart migration mapped one major brand's sizes incorrectly, so customers ordered the wrong size.

FIX

- Correct the mapping; notify affected customers; easy exchanges.

MEASURE RECOVERY

- Return rate by brand and reason.

PREVENT RECURRENCE

- Validate catalogue migrations against sample products before launch.

4. Common mistakes

- Blaming the seller's quality without checking reasons.

> **60-second answer**
>
> Returns were mostly sizing-related after a size-chart migration; one major brand's sizes were mapped wrong. Fix the mapping, support exchanges and validate catalogue migrations.

Q21 · Video upload completion down 25%

Creator platform (YouTube Shorts-style) · Metric: Uploads completed ÷ started · Medium

Prompt

Completed uploads from creators fell 25% over five days. Diagnose.

1. Clarifying questions to ask

- Started vs completed? App or web?

2. Hypothesis buckets

- App bug.
- Upload infrastructure change.
- Policy or copyright checks.
- Network issues.

Interviewer reveals (share only when asked)

- Failures are on uploads larger than 100 MB.
- A CDN configuration change reduced the upload timeout.

3. Model investigation path

1. Split by file size → large files fail.
2. Infra changes → timeout reduced.

Root cause

An infrastructure change reduced upload timeouts, causing large videos to fail.

FIX

- Restore timeout; use resumable chunked uploads.

MEASURE RECOVERY

- Upload success by size band, retries.

PREVENT RECURRENCE

- Synthetic large-file upload tests after infra changes.

4. Common mistakes

- Assuming creators lost motivation.

60-second answer

Only uploads over 100 MB failed after a CDN change shortened timeouts. Restore the timeout, add resumable uploads and test large files after infrastructure changes.

Q22 · Ad click-through rate down 15%

Ads platform · Metric: Ad CTR · Hard

Prompt

Ad CTR on the feed fell 15% while impressions rose. What happened?

1. Clarifying questions to ask

- Which placements, advertisers, devices?

2. Hypothesis buckets

- Ad quality drop.
- More impressions at worse positions.
- Bot traffic.
- Tracking issue.

Interviewer reveals (share only when asked)

- Clicks are flat; impressions rose 18%.
- Ad load per session was increased last week.
- Lower-position ads have low CTR.

3. Model investigation path

1. Clicks flat, impressions up → denominator effect.
2. Ad load change → extra lower-quality slots.

Root cause

Increasing ad load added low-position impressions, diluting CTR while clicks stayed flat.

FIX

- Evaluate on revenue, advertiser ROI and user experience, not CTR alone; tune ad load.

MEASURE RECOVERY

- Revenue per session, advertiser conversions, session length, user complaints.

PREVENT RECURRENCE

- Ad-load experiments with user guardrails.

4. Common mistakes

- Treating CTR decline as ad quality decline without checking impressions.

> **60-second answer**
>
> Clicks were flat and impressions rose because ad load increased. CTR diluted mathematically. Judge ad load on revenue, advertiser ROI and user experience guardrails.

Q23 · Group creation down in a community product

Community feature (Facebook Groups-style) · Metric: New active groups · Hard

Prompt

New groups created per week fell 20%. Find the cause.

1. Clarifying questions to ask

- Groups created or groups that become active?

2. Hypothesis buckets

- Entry point removed.
- Spam enforcement change.
- Competition.
- Seasonality.

Interviewer reveals (share only when asked)

- Group creation from the main navigation is unchanged.
- Creation from the 'events' entry point dropped to near zero after a navigation redesign removed that entry point.

3. Model investigation path

1. Split by entry point → events entry point vanished.
2. Navigation redesign removed it.

Root cause

A navigation redesign removed a high-volume entry point for creating groups.

FIX

- Restore or replace the entry point; measure group activation, not just creation.

MEASURE RECOVERY

- Groups created by entry point; groups active after 14 days.

PREVENT RECURRENCE

- Check traffic per entry point before removing navigation elements.

4. Common mistakes

- Ignoring entry-point segmentation.

> **60-second answer**
>
> Creation from main navigation was unchanged; the events entry point disappeared in a redesign. Restore it and measure activation of new groups, not just creation.

Q24 · Time-to-resolution up for B2B support

B2B SaaS support · Metric: Median time to resolution · Medium

Prompt

Median support ticket resolution time rose from 6 hours to 11 hours. Diagnose.

1. Clarifying questions to ask

- Median or mean? All tiers?

2. Hypothesis buckets

- Higher volume.
- More complex issues.
- Staffing changes.
- Routing changes.

Interviewer reveals (share only when asked)

- Ticket volume is flat.
- Tickets wait longer before first response.
- A new routing rule sends all enterprise tickets to one small team.

3. Model investigation path

1. Volume flat → not demand.
2. First response time up → queueing.
3. Routing rule overloaded one team.

Root cause

A routing rule concentrated enterprise tickets on a small team, creating a queue.

FIX

- Rebalance routing; add overflow rules.

MEASURE RECOVERY

- First response time, queue length by team, resolution time by tier.

PREVENT RECURRENCE

- Simulate routing changes on historical ticket data.

4. Common mistakes

- Hiring more agents without fixing routing.

> **60-second answer**
>
> Volume was flat but first-response time rose because a routing rule sent all enterprise tickets to one small team. Rebalance routing, add overflow and simulate routing changes beforehand.